

Supplementary material to ‘Analytical Estimate of DEM Accuracy (MSE) as a Function of Cell Size and Dataset Elevation Error’

Appendix 1

We show here how to estimate the a_j term, the first component of the bound in the bilinear interpolant case. Although, for small h , the other contributions can be neglected, we nevertheless provide their analytical expressions for a_j , b_j and c_j .

$$a_j = \frac{1}{2} \left[s_j (1 - s_j) \frac{\partial^2 Z}{\partial x^2} + t_j (1 - t_j) \frac{\partial^2 Z}{\partial y^2} \right] \quad (1)$$

$$b_j = \frac{1}{6} \left[s_j (2s_j^2 - 3s_j + 1) \frac{\partial^3 Z}{\partial x^3} + t_j (2t_j^2 - 3t_j + 1) \frac{\partial^3 Z}{\partial y^3} \right] \quad (2)$$

$$c_j = \frac{1}{24} \left[s_j (-3s_j^3 + 6s_j^2 - 4s_j + 1) \frac{\partial^4 Z}{\partial x^4} + t_j (-3t_j^3 + 6t_j^2 - 4t_j + 1) \frac{\partial^4 Z}{\partial y^4} \right] \quad (3)$$

Since the a_j term involves second order derivative estimates the stencil presented in Fig. 1 should be conveniently expanded to include some neighbours. See Fig. 2 for the schema.

Let $B(x,y)$ denote the bilinear interpolant based on the set $\{2,4,6,8\}$. It can be shown that, among other options, the required partial derivatives can be estimated as

$$\begin{aligned} \left. \frac{\partial^2 Z}{\partial x^2} \right|_{(x_j, y_j)} &\approx 2 \frac{(B(x_3, y_3) - Z_3)}{h^2} \\ \left. \frac{\partial^2 Z}{\partial y^2} \right|_{(x_j, y_j)} &\approx 2 \frac{(B(x_5, y_5) - Z_5)}{h^2} \end{aligned} \quad (4)$$

These expressions are not the only available alternatives, but they are simple and quick to evaluate.

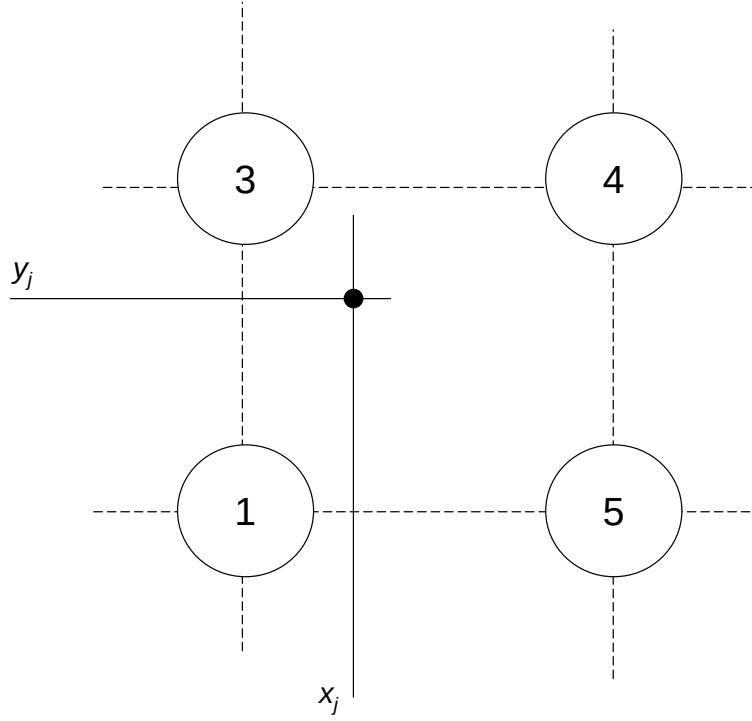


Figure 1 Neighbourhood of the j -th control point, involving points of the regular grid.

Appendix 2

Here we analyze the popular nearest-neighbour (NN) interpolant in order to establish a functional relationship between DEM accuracy and cell size. NN is exact only for constant functions; thus, as an interpolant, it is of first order. As before, we assume that the j -th control point lies within a cell whose vertices are labelled $\{1,3,4,5\}$ (see Figure 1) the NN expression is simply

$$H(x_j, y_j) = \begin{cases} Z_1 & \text{if } 0 \leq x_j \leq h/2 \text{ and } 0 \leq y_j \leq h/2 \\ Z_3 & \text{if } 0 \leq x_j \leq h/2 \text{ and } h/2 < y_j \leq h \\ Z_5 & \text{if } h/2 < x_j \leq h \text{ and } 0 \leq y_j \leq h/2 \\ Z_4 & \text{if } h/2 < x_j \leq h \text{ and } h/2 < y_j \leq h \end{cases} \quad (5)$$

where $(x_j, y_j) \in [0, h]^2$. The Taylor expansion around (x_j, y_j) is presented in Eq. 6.

$$\begin{aligned} Z(x, y) &= Z(x_j, y_j) + \frac{\partial Z}{\partial x}(x - x_j) + \frac{\partial Z}{\partial y}(y - y_j) + \dots \\ &\quad \frac{1}{2} \frac{\partial^2 Z}{\partial x^2} (x - x_j)^2 + \frac{\partial^2 Z}{\partial x \partial y} (x - x_j)(y - y_j) + \frac{1}{2} \frac{\partial^2 Z}{\partial y^2} (y - y_j)^2 + \dots \end{aligned} \quad (6)$$

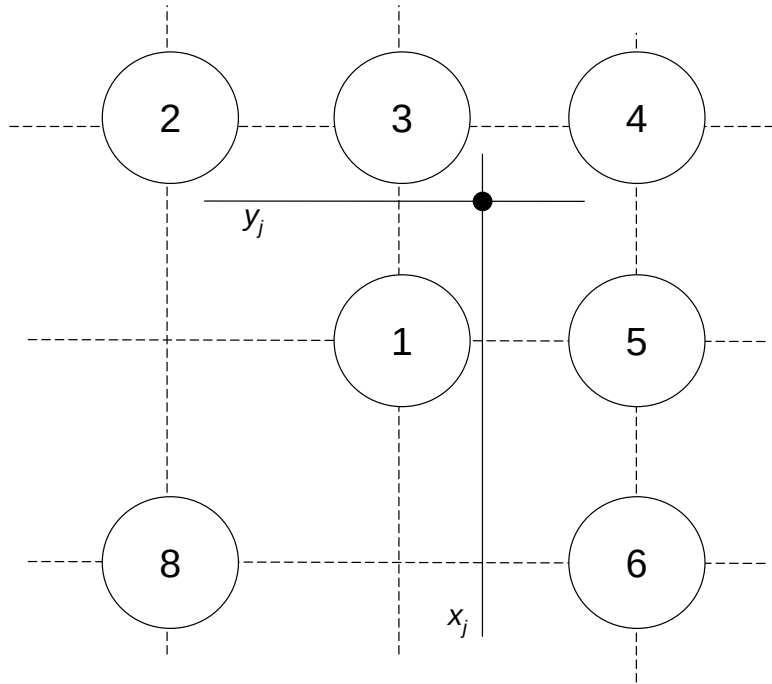


Figure 2 Expanded neighbourhood of the j -th control point, involving extra points of the regular grid.

After changing $(x_j, y_j) = h^*(s_j, t_j)$ it can be shown that (for $(s_j, t_j) \in [0, 1/2]^2$)

$$H(\mathbf{Z}, h)|_{(x_j, y_j)} = Z|_{(x_j, y_j)} + \frac{h}{2} \left(s_j \frac{\partial Z}{\partial x} + t_j \frac{\partial Z}{\partial y} \right) + O(h^2) \quad (7)$$

The other quadrants will produce a similar expression. Since $Z = z + e$ it follows that, for the j -th control point,

$$\begin{aligned} H(\mathbf{Z}, h) &= z|_{(x_j, y_j)} + e|_{(x_j, y_j)} + \frac{h}{2} \left(s_j \frac{\partial Z}{\partial x} + t_j \frac{\partial Z}{\partial y} \right) + O(h^2) = \\ &= z|_{(x_j, y_j)} + e|_{(x_j, y_j)} + ha|_{(x_j, y_j)} + h^2 b|_{(x_j, y_j)} + O(h^3) \end{aligned} \quad (8)$$

For short we define $a|_{(x_j, y_j)}$ and $b|_{(x_j, y_j)}$ as

$$a|_{(x_j, y_j)} = \frac{1}{2} \left(s_j \frac{\partial Z}{\partial x} + t_j \frac{\partial Z}{\partial y} \right); b|_{(x_j, y_j)} = \frac{1}{2} \frac{\partial^2 Z}{\partial x^2} s_j^2 + \frac{\partial^2 Z}{\partial x \partial y} s_j t_j + \frac{1}{2} \frac{\partial^2 Z}{\partial y^2} t_j^2 \quad (9)$$

Collecting values at the j -th control point, we define the vectors $\mathbf{e}$, $\mathbf{a}$, and $\mathbf{b}$ as before. After some straightforward manipulations—squaring both sides and averaging over all control points—we obtain the following concise relationship

$$\frac{1}{\#CP} \sum_{all\ CP} \left(H(x_j, y_j) - z|_{(x_j, y_j)} \right)^2 \leq \frac{\mathbf{e}^T \mathbf{e}}{\#CP} + 2 \frac{|\mathbf{e}^T \mathbf{a}|}{\#CP} h + 2 \frac{|\mathbf{a}^T \mathbf{a} + \mathbf{e}^T \mathbf{b}|}{\#CP} h^2 + O(h^3) \quad (10)$$

The left-hand side represents the desired MSE. The right-hand side consists of three dominant terms. The first two are affected by dataset accuracy, while the third depends on the cell size and certain partial derivatives of the topographic surface. The first and second terms vanish when the dataset elevations are exact. The first term is equivalent to dataset accuracy. The second term involves a weighted average of the dataset error and the first partial derivatives of the true surface. The elements of vectors $\mathbf{a}$ and $\mathbf{b}$ include partial derivative values at the control points. The first-order derivatives can be estimated from the available elevations as

$$\frac{\partial Z}{\partial x} \approx \frac{Z_5 - Z_1}{h}; \frac{\partial Z}{\partial y} \approx \frac{Z_3 - Z_1}{h} \quad (11)$$

The second-order derivatives can be estimated using the bilinear interpolator, as shown in Appendix 1. Summing up, and following an approach analogous to the bilinear interpolation case, we obtain an analytical expression linking the model MSE to the cell size. It is

$$E_{\text{model}} \leq E_{\text{data}} + C_1 h^p + C_2 h^{2p}; p = 1 \quad (12)$$

In this case, p is just 1. The coefficients C_1 and C_2 are now a function related with the first and second partial derivatives thus confirming in part the conjecture posed by Li (1992).